CSCE 411: Design and Analysis of Algorithms (Honors)

Week 7, Exam 2 Review

Date: October 15, 2026 *Nate Veldt, updated by Samson Zhou*

1 Test format and instructions

- Some multiple choice questions, some long answer questions
- You do not need to prove something unless you are explicitly asked to prove something
- You may be asked to explicitly prove something
- Scratch paper will be provided
- No calculators or electronic devices. You will not need them. If you have to do any computations, they will be basic enough to do by hand.
- You will put your name on the front page, and put your initials at the top of every other page.

A few other notes:

- If something was on a homework or in a lecture, it could be on the test.
- Some questions may involve a combination of a couple different things we've seen in different lectures.
- If we covered a proof of a result, you can reference it as fact without re-proving it, unless the question specifically is asking you to provide proof details.

2 Topics Covered on the Test

2.1 Summary of the summary

- Graph basics (adjacency matrix, adjacency list, directed/undirected, weighted/unweighted, cyclic/acyclic)
- (BFS) Breadth-first search
- (DFS) Depth-first search
- (SSSP) Single source shortest paths (Bellman-Ford, Dijkstra's)
- Maximum s-t flow and minimum s-t cut (Ford-Fulkerson, Edmonds-Karp)
- Linear programming, duality
- Continuous optimization (Convexity, smoothness, gradient descent)

For all of the graph algorithms, here's the short answer on what you should know:

- How is the problem defined? In particular, what type of graph does it apply to? Are there restrictions?
- What algorithms solve the problem, what is their basic mechanism, and their runtime?
- Don't just know how to eye-ball a solution for a small graph. You should know the mechanics of each algorithm well enough that you could apply it to a small graph, could write out pseudocode if need be, and provide a runtime analysis

2.2 Graph Basics

- Representing a graph as a list or a matrix
- Undirected and directed, weighted and unweighted (which of these can be viewed as special cases of others?)
- Strongly connected and weakly connected components. What is their relation? How are they defined?
- What is a bipartite graph?

2.3 Breadth first search specifics

- Running a BFS on a weighted graph just means ignore the edge weights. (You wouldn't say "BFS doesn't apply if the graph is weighted.")
- Remember what a breadth-first tree is and how to find it.
- Make sure you know the runtime

2.4 Depth-first search

- Know the different edge types in a depth-first search and what they mean
- Know the two main applications we discussed in class and how to use DFS for them.

2.5 Single source shortest paths algorithm

- Two algorithms we saw: Bellman-Ford, Dijkstra's
- You should understand the basic mechanism behind each, and different pros/cons of their runtimes

2.6 Min-cut Max-Flow Problem and algorithms

- What is the definition of an s-t flow function? What is an s-t cut in a directed graph?
- What is the basic idea behind the algorithms we have seen? Hint: it isn't that you repeatedly try to send flow from s to t as long as you can find unsaturated paths in the original graph G.
- What is an augmenting path and a residual graph?
- What is the basic mechanism behind Ford-Fulkerson? What's its runtime and why?
- What is the difference between Ford-Fulkerson and Edmonds-Karp? What is the runtime difference?
- You would not be expected to reproduce a long proof such as the one we did for Edmond's Karp, but you might be expected to understand/explain one small subset.
- You should know that what the maximum-flow minimum-cut theorem says and how to use it to verify that a flow or cut is optimal or not optimal.

2.7 Linear programming and duality

- What is a linear program? Identify its decision variables, objective function, and constraints.
- How do you formulate a word problem as a linear program? You should be able to explain what each variable and constraint represents.
- What does it mean for a solution to be feasible or optimal? What does it mean for a linear program to be infeasible or unbounded?
- What does the feasible region look like geometrically? You should be able to solve a small linear program with two variables by drawing its constraints and comparing objective values.
- Given a primal linear program, how do you write its dual? Pay attention to the directions of inequalities and restrictions on the signs of variables.
- What does weak duality say? You should understand its proof and explain where primal and dual feasibility are used.
- What does strong duality say when the primal has a finite optimal value? How can a primal feasible solution and a dual feasible solution with equal objective values certify that both are optimal?
- How is the maximum-flow minimum-cut theorem related to linear programming duality?

2.8 Convex optimization and gradient descent

- What does it mean for a set or a function to be convex? What is the first-order characterization of convexity for a differentiable function?
- What is the relationship between local and global minima of a convex function? For differentiable, unconstrained convex minimization, why does a zero gradient certify a global minimum?
- What is a gradient, and how do you compute it? How does it relate to the derivative of a function of one variable?
- What does it mean for a function to be Lipschitz continuous or smooth? Distinguish a bound on changes in function values, a bound on the gradient itself, and a bound on changes in the gradient.

- What is the basic mechanism behind gradient descent? Know the update rule

$$x_{t+1} = x_t - \eta \nabla f(x_t),$$

  where $\eta > 0$ is the step size. You should be able to write pseudocode and carry out a few iterations on a simple function.

- What role does the step size play? How does smoothness help us choose it, and why can a step size that is too large cause oscillation or divergence?

- How does the smoothness inequality show that a suitable gradient descent step decreases the objective when the gradient is nonzero?

3 Practice Problems

1. When does breadth-first search solve the same problem as Dijkstra's algorithm?
2. What are applications of a breadth first search? What problem(s) does it solve?
3. What are applications for depth first search? What problem(s) does it solve?
4. Write down the adjacency matrix and the adjacency list for each of the sample graphs.
5. Draw the breadth-first tree of the sample graph 1 from starting node e
6. Find the minimum s-t cut of sample graph 1 when $s = a$ and $t = f$. Explain whether you get the same minimum s-t cut value when $s = f$ and $t = a$.
7. Write out a valid s-t flow for the graph in Figure 2.
8. Solve other BFS, DFS, single source shortest path problems on the sample graphs (e.g., try a different source and/or sink node, change weights slightly, etc.)

Sample Graphs

Figure 1: Sample graph 1

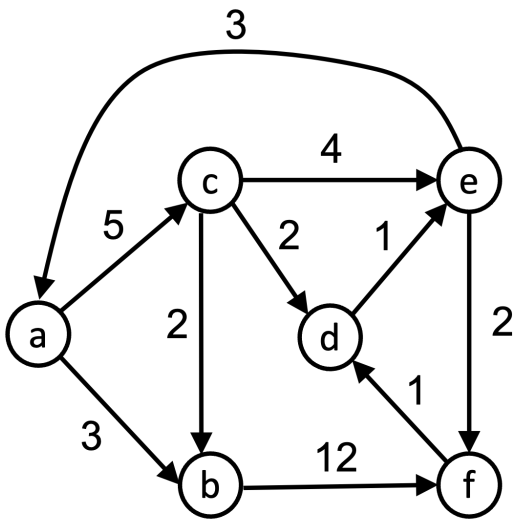


Figure 2: Sample graph 2

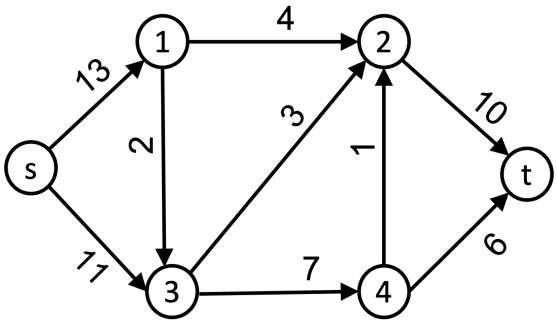
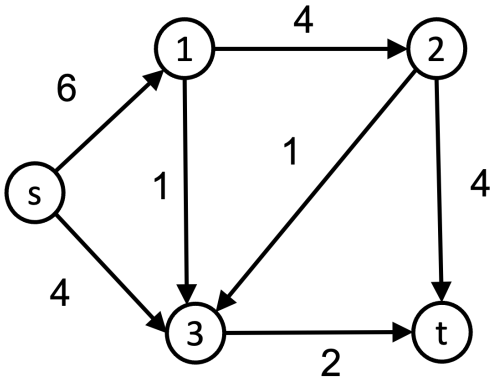


Figure 3: Sample graph 3



9. Formulate each of the following problems as a linear program. Clearly define your variables, objective function, and constraints. Amounts may be fractional.

   (a) A factory produces chemicals A and B. Each unit of A requires two hours on machine 1 and one hour on machine 2, and earns a profit of \$3. Each unit of B requires one hour on each machine and earns a profit of \$2. Machine 1 is available for six hours and machine 2 is available for four hours. Maximize the total profit.

   (b) Each unit of food A costs \$3 and provides two units of protein and one unit of fiber. Each unit of food B costs \$3 and provides one unit of protein and two units of fiber. Find a minimum-cost combination providing at least six units of protein and at least six units of fiber.

10. Consider the linear program

$$\begin{aligned} \text{maximize} \quad & 3x + 2y \\ \text{subject to} \quad & x + y \leq 4, \\ & 2x + y \leq 6, \\ & x, y \geq 0. \end{aligned}$$

    (a) Draw the feasible region and identify its vertices. Find an optimal solution and its objective value.

    (b) Write the dual linear program.

    (c) Find a dual feasible solution that certifies your primal solution is optimal. Explain why this is a valid certificate.

11. Consider the primal and dual linear programs

$$\begin{array}{lll} \text{maximize} & c^\mathsf{T} x & \\ \text{subject to} & Ax \leq b, \quad x \geq 0 & \end{array} \qquad \begin{array}{lll} \text{minimize} & b^\mathsf{T} y & \\ \text{subject to} & A^\mathsf{T} y \geq c, \quad y \geq 0. & \end{array}$$

    Recall that $y^\mathsf{T} A x = (A^\mathsf{T} y)^\mathsf{T} x$. Prove that every primal feasible x and dual feasible y satisfy

$$c^\mathsf{T} x \leq b^\mathsf{T} y.$$

    Explain where each feasibility condition is used. Do not invoke strong duality.

12. Classify each linear program as infeasible, feasible with a finite optimum, or feasible with an unbounded objective. For each feasible program, also determine whether its feasible region is bounded. Justify your answers.

    (a) Maximize $x + y$ subject to $x + y \leq 1$, $x + y \geq 2$, and $x, y \geq 0$.

    (b) Maximize $x + y$ subject to $x - y \leq 1$ and $x, y \geq 0$.

    (c) Minimize $x + y$ subject to $x + y \geq 1$ and $x, y \geq 0$.

    (d) Maximize x subject to $x \leq 2$ and $x, y \geq 0$.

13. Let $G = (V, E)$ be a directed graph with source s, sink t, and nonnegative edge capacities c_e.

    (a) Formulate maximum s-t flow as a linear program. Specify the variables, capacity constraints, flow conservation constraints, and objective function.

    (b) Given an s-t cut, explain why its capacity is an upper bound on the value of every feasible flow. How can a flow and a cut together certify optimality?

14. Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex.

    (a) State the first-order inequality that characterizes convexity.

    (b) Use this inequality to prove that if $\nabla f(z) = 0$, then z is a global minimizer.

    (c) Give a differentiable, nonconvex function with a point where its derivative is zero but which is not a local minimum.

15. For each function and domain below, determine whether the function is convex, Lipschitz continuous, and smooth (has a Lipschitz continuous derivative). Justify your answers and give valid Lipschitz and smoothness constants whenever they exist.

    (a) $f(x) = x^2$ on $\mathbb{R}$.

    (b) $f(x) = x^2$ on $[-2, 2]$.

    (c) $f(x) = |x|$ on $\mathbb{R}$.

16. Consider gradient descent on $f(x) = (x - 3)^2$, starting from $x_0 = 1$ and using a constant step size $\eta > 0$.

(a) Compute x_1, x_2, x_3 for $\eta = 1/4$. Repeat for $\eta = 1/2$.

(b) Express $x_{t+1} - 3$ in terms of $x_t - 3$ and η. Use this recurrence to determine all positive step sizes for which the iterates converge to the minimizer from $x_0 = 1$.

(c) Describe what happens for $\eta = 1$ and for $\eta = 3/2$. Does the objective value decrease at every iteration?

17. Consider $f(x_1, x_2) = x_1^2 + 2x_2^2$.

(a) Compute ∇f and identify a global minimizer.

(b) Starting from $x^{(0)} = (4, 2)$, carry out two iterations of gradient descent with step size $\eta = 1/4$. Compute the objective value at the initial point and at each new iterate.

(c) Give a valid smoothness constant L with respect to the Euclidean norm and justify it. How does the chosen step size compare with $1/L$?